Roll No.: ....

END SEMESTER EXAMINATION JUNE 2025

Name of the Program: B.Tech | Semester: VI

Name of the Course: Big Data Storage and Processing | Course Code: TCS671

Time: 3 Hours | **Maximum Marks: 100**

Note:

i. **All questions are compulsory.**
ii. Answer any two sub questions among a, b & c in each main questions
iii. Total marks in each main question are twenty.
iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	Explain the 5V characteristics of Big Data with real-world examples. Discuss technological challenges in managing petabyte-scale datasets.	CO1
(b)	Compare traditional data architecture with Big Data architecture. Discuss the classification of digital data in detail.	CO1
(c)	Discuss the key steps in Capturing Big Data. What are different data collection techniques used in big data? Discuss in detail.	CO1
Q2		
(a)	Describe HDFS architecture with a neat diagram. Explain how data replication ensures fault tolerance in distributed systems.	CO2
(b)	Discuss the data flow of reading and writing the data from HDFS with help of diagrams.	CO2
(c)	Discuss key components of Hadoop CLI. What are the different limitations of Hadoop? Discuss in detail.	CO2
Q3		
(a)	Explain the complete workflow of MapReduce with shuffle-sort phase diagram. How does speculative execution improve job performance?	CO4
(b)	Explain how Hadoop ensures data integrity, covering the mechanisms involved from data input to storage. Discuss the role of checksums in maintaining data integrity within HDFS.	CO3 CO3
(c)	What is serialization and its importance in Hadoop? Explain the task execution process in Hadoop with diagram.	
Q4		
(a)	Create a step-by-step guide to configure 5-node Hadoop cluster with hardware/software specifications and network topology.	CO4 CO5
(b)	Differentiate between standalone, pseudo-distributed and fully-distributed modes. What hardware considerations are vital for production clusters?	CO4
(c)	Compare between JobTracker and TaskTracker in detail. Discuss the use case of different modes of installations of Hadoop.	
Q5		
(a)	Compare YARN schedulers (Fair vs Capacity) with resource allocation diagrams. How does ZooKeeper ensure cluster coordination?	CO5
(b)	Prove CAP theorem with NoSQL examples. Design a column-family database schema for e-commerce product catalog.	CO6
(c)	Discuss NoSQL architecture with diagram. Compare RDBMS vs NoSQL in detail.	CO6